

ECN 594: Practice Midterm Exam - SOLUTIONS

1. Short Answer Questions (30 points)

1. (a) (4 points) Profit-maximizing price:

$$\frac{P-MC}{P} = \frac{1}{|\varepsilon|} = \frac{1}{4} \implies P - 15 = 0.25P \implies 0.75P = 15 \implies P^* = \$20$$

- (b) (4 points) Interpretation of ξ_{jt} :

ξ_{jt} is unobserved product quality—factors affecting demand that the researcher doesn't observe (e.g., brand reputation, advertising, design).

- (c) (4 points) Demographics fully eliminate IIA:

False. IIA still holds within demographic groups. Only mixed logit (random coefficients) fully relaxes IIA.

- (d) (4 points) Cost shifters as instruments:

Relevance: Higher input costs $\rightarrow$ higher prices. Exogeneity: Input prices determined by upstream markets, not correlated with local demand shocks ξ_{jt} .

- (e) (4 points) Cross-price elasticity depends on similarity:

False. In logit, $\eta_{jk} = -\alpha p_k s_k$ —depends only on k 's price and share, not on similarity between j and k . This is the IIA limitation.

- (f) (3 points) MC pricing gives zero profit:

False. $\pi = pq - F - cq = cq - F - cq = -F < 0$. MC pricing requires government subsidies.

- (g) (4 points) False CS formula:

Incorrect. Correct formula is $CS = \frac{1}{|\alpha|} \ln(1 + e^{\delta_1} + e^{\delta_2})$. The proposed formula is missing: (1) the log, (2) the $1/|\alpha|$ scaling, (3) the outside option term.

- (h) (3 points) High-type IC always binds:

True. At optimum, high type is just indifferent between high and low product. If slack, firm could raise high-type price.

2. Demand Estimation (30 points)

2. (a) (6 points) Define δ_{jt} and derive Berry inversion:

$$\delta_{jt} = \beta \cdot \text{screen}_j + \alpha \cdot p_{jt} + \xi_{jt} \text{ (part of utility common to all consumers).}$$

$$\text{Berry inversion: } \delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$$

- (b) (6 points) Regression equation:

$$\ln(s_{jt}) - \ln(s_{0t}) = \beta \cdot \text{screen}_j + \alpha \cdot p_{jt} + \xi_{jt}$$

Dependent variable: $\ln(s_{jt}) - \ln(s_{0t})$. Regressors: screen_j, p_{jt} . Error: ξ_{jt} .

- (c) (6 points) OLS bias:

$\text{Cov}(p_{jt}, \xi_{jt}) > 0$: high-quality phones command higher prices.

Upward bias (less negative $\hat{\alpha}$): OLS attributes high demand for expensive phones to low price sensitivity rather than unobserved quality.

- (d) (6 points) Cost shifters exclusion restriction:

If chip prices affect product quality (e.g., better chips $\rightarrow$ faster phones $\rightarrow$ higher ξ), then chip prices are correlated with ξ_{jt} , violating exclusion.

- (e) (6 points) Advantage of demographic interactions:

Allows heterogeneous preferences across consumers, partially addressing IIA: older vs. younger consumers may prefer different screen sizes, generating more realistic substitution patterns.

3. Two-Part Tariffs (20 points)

3. (a) (10 points) Profit as a function of F and p :

$q = 30 - 3p$, choke price $p = 10$.

$CS(p) = \frac{1}{2}(30 - 3p)(10 - p) = \frac{3}{2}(10 - p)^2$, so max fee $F = \frac{3}{2}(10 - p)^2$.

$\Pi = 200 \left[\frac{3}{2}(10 - p)^2 + (p - 1)(30 - 3p) \right]$

- (b) (10 points) Show $p^* = c = 1$:

Per-consumer profit: $\pi(p) = \frac{3}{2}(10 - p)^2 + (p - 1)(30 - 3p)$

$\frac{d\pi}{dp} = -3(10 - p) + (30 - 3p) - 3(p - 1) = -30 + 3p + 30 - 3p - 3p + 3 = 3 - 3p$

$3 - 3p = 0 \Rightarrow p^* = 1 = c$

4. Self-Selection (20 points)

4. (a) (6 points) Perfect price discrimination:

Corporate: $P_B = \$800$. Leisure: $P_E = \$150$.

Profit = $50(800 - 50) + 100(150 - 50) = 50(750) + 100(100) = 37,500 + 10,000 = \$47,500$

- (b) (6 points) IC constraint for corporate travelers:

$800 - P_B \geq 300 - P_E \implies P_B - P_E \leq 500$

- (c) (8 points) Self-selection prices:

Binding: Leisure IR ($P_E = 150$) and Corporate IC ($P_B = P_E + 500 = 650$).

Check leisure IC: $150 - 150 = 0 \geq 200 - 650 = -450 \checkmark$

Profit = $50(650 - 50) + 100(150 - 50) = 50(600) + 100(100) = 30,000 + 10,000 = \$40,000$